

Maharashtra Public Service Commission Assessment

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Assistant Professor Statistics Group A Higher and Technical Education Department

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Assistant Professor Statistics Group A Higher and Technical Education Department

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Sub-Section Number :	1
Sub-Section Id :	4410092998255
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 1 Question Id : 4410092088947 Question Type : MCQ Option Shuffling : No

Suppose (a_n) is a bounded sequence in $\mathbb{R}$. Which of the following must be true according to the Bolzano–Weierstrass theorem?

Options :

1. The sequence (a_n) is monotone and bounded.
2. There exists at least one convergent subsequence of (a_n) .
3. Every subsequence of (a_n) converges to the same limit.
4. The sequence (a_n) itself must converge.

Question Number : 2 Question Id : 4410092088747 Question Type : MCQ Option Shuffling : No

For which of the following sets of numbers does the greatest lower bound (infimum) fail to exist within the set itself?

Options :

1. The set of all numbers formed by dividing one by every possible positive whole number.
2. The set of all numbers formed by subtracting one from every possible positive whole number.
3. The set of all numbers formed by multiplying five by every possible positive whole number.
4. The set of all numbers formed by multiplying every possible positive whole number by itself.

Question Number : 3 Question Id : 4410092088749 Question Type : MCQ Option Shuffling : No

Which of the following is a necessary property for a subset W of a vector space V to be a subspace?

Options :

1. W must be non-empty
2. W must contain the zero vector of V
3. W must contain every vector of V
4. W must contain at least one nonzero vector

Question Number : 4 Question Id : 4410092088750 Question Type : MCQ Option Shuffling : No

In a linear system, if adjoining the constant vector to the coefficient matrix strictly increases its overall rank, what is the definitive geometric consequence for the transformation?

Options :

1. The target vector lies outside the column space, guaranteeing that no solution can exist.
2. The target vector lies outside the row space, guaranteeing that infinite solutions will exist.
3. The target vector lies inside the column space, guaranteeing that one unique solution exists.
4. The target vector lies inside the row space, guaranteeing that no solution can exist.

Question Number : 5 Question Id : 4410092108516 Question Type : MCQ Option Shuffling : No

Given the quadratic form $Q(x, y, z) = x^2 + 2y^2 + 3z^2 + 2xy + 2xz + 2yz$, what is the rank of the associated symmetric matrix?

Options :

1. The rank is 3 because the matrix is full rank and invertible.
2. The rank is 0 because all entries in the matrix are zero.
3. The rank is 1 because the matrix is not invertible.
4. The rank is 2 because one row is a linear combination of the others.

Question Number : 6 Question Id : 4410092108620 Question Type : MCQ Option Shuffling : No

Which expression correctly represents the Newton-Raphson iteration for a function $f(x)$?

Options :

1. $x_{\text{new}} = x_{\text{old}} - f(x_{\text{old}}) / f'(x_{\text{old}})$
2. $x_{\text{new}} = x_{\text{old}} + f(x_{\text{old}}) / f(x_{\text{old}})$
3. $x_{\text{new}} = x_{\text{old}} - f(x_{\text{old}}) / f(x_{\text{old}})$
4. $x_{\text{new}} = x_{\text{old}} + f(x_{\text{old}}) / f'(x_{\text{old}})$

Question Number : 7 Question Id : 4410092108517 Question Type : MCQ Option Shuffling : No

If the Newton–Raphson method is used on $f(x) = x^3 - 2x + 2$ starting from $x_0 = 0$, what is the main cause if the sequence oscillates without convergence?

Options :

1. The derivative at iterates alternates in sign causing overshooting
2. The initial approximation lies in a region where tangent updates cycle instead of converge
3. The function exhibits smooth continuity, but Newton steps misalign due to slope reversals
4. The iteration step magnitude remains bounded, preventing escape from the oscillatory loop

Question Number : 8 Question Id : 4410092088756 Question Type : MCQ Option Shuffling : No

When displaying the empirical frequency distribution of a continuous quantitative variable, which standard graphical method is fundamentally required to appropriately represent grouped class intervals?

Options :

1. A histogram, utilizing contiguous vertical bars to display continuous numerical data.
2. A pie chart, utilizing partitioned circular sectors to display discrete categorical data.
3. A scatter plot, utilizing mapped Cartesian coordinates to display paired bivariate data.
4. A line graph, utilizing sequentially connected points to display ordered longitudinal data.

Question Number : 9 Question Id : 4410092088755 Question Type : MCQ Option Shuffling : No

If the mean deviation of a dataset is zero, what does this indicate about the data values?

Options :

1. Data are highly dispersed
2. Data are symmetrically distributed
3. Quartiles are equal
4. All values are equal to the mean

Question Number : 10 Question Id : 4410092108523 Question Type : MCQ Option Shuffling : No

A researcher measures the coefficient of variation for three experimental groups and finds values of 0.18, 0.22, and 0.25, respectively. If a fourth group yields a mean that is double that of the first group and a standard deviation that is half that of the first group, what is the coefficient of variation for the fourth group?

Options :

1. 0.045
2. 0.09
3. 0.36
4. 0.72

Question Number : 11 Question Id : 4410092108622 Question Type : MCQ Option Shuffling : No

The mean of n observations is $\bar{x}$. If one observation x is removed, the new mean is:

Options :

1. $\frac{n\bar{x} + x}{n - 1}$
2. $\frac{n\bar{x} - x}{n - 1}$
3. $\frac{n\bar{x}}{n - 1}$
4. $\frac{\bar{x} - x}{n}$

Question Number : 12 Question Id : 4410092108623 Question Type : MCQ Option Shuffling : No

Consider two groups:

A: $n_1 = 25$, $\bar{x}_1 = 50$, $\sigma_1^2 = 16$

B: $n_2 = 25$, $\bar{x}_2 = 60$, $\sigma_2^2 = 16$

If all values in Group A are multiplied by 3 and values in Group B are reduced by 10, then combined mean and combined variance are, respectively:

Options :

1. 100 and 2580
2. 100 and 2000
3. 60 and 2500
4. 60 and 2000

Question Number : 13 Question Id : 4410092088759 Question Type : MCQ Option Shuffling : No

Which aspect of scatter diagrams can help detect heteroscedasticity in a data set?

Options :

1. The points display a perfectly constant spread around the trend line.
2. The points display a noticeably changing spread around the trend line.
3. The points display a completely circular pattern around the center mass.
4. The points display a strictly linear alignment without any visual spread.

Question Number : 14 Question Id : 4410092088760 Question Type : MCQ Option Shuffling : No

In statistical modeling, what fundamental theoretical property qualifies the Spearman rank correlation coefficient as a strictly non-parametric measure of association?

Options :

1. It fundamentally analyzes the ordinal rankings of data, entirely bypassing requirements for specific distributional parameters.
2. It fundamentally analyzes the exact magnitudes of data, entirely bypassing requirements for specific distributional parameters.
3. It fundamentally analyzes the ordinal rankings of data, strictly mandating requirements for specific distributional parameters.
4. It fundamentally analyzes the exact magnitudes of data, strictly mandating requirements for specific distributional parameters.

Question Number : 15 Question Id : 4410092088761 Question Type : MCQ Option Shuffling : No

For an experiment where a spinner is spun and can land on any real number between 0 and 1, how is the sample space best described?

Options :

1. Only the numbers 0 and 1
2. All positive integers
3. All integers from 0 to 1
4. All real numbers between 0 and 1

Question Number : 16 Question Id : 4410092108528 Question Type : MCQ Option Shuffling : No

If prior probabilities for two events A and B are 0.4 and 0.6, and the likelihoods $P(E|A) = 0.7$ and $P(E|B) = 0.2$, what is the posterior probability of A given E?

Options :

1. 0.4
2. 0.7
3. 0.2
4. 0.3

Question Number : 17 Question Id : 4410092108537 Question Type : MCQ Option Shuffling : No

A sequence of random variables X_n converges in distribution to a random variable Z. Which of the following properties holds for every bounded continuous function f?

Options :

1. For every x, the probability that X_n equals x converges to the probability that Z equals x.
2. The expected value of $f(X_n)$ converges to the expected value of $f(Z)$.
3. The sequence X_n converges almost surely to Z for every bounded function.
4. The variance of X_n converges to the variance of Z for every continuous function.

Question Number : 18 Question Id : 4410092088775 Question Type : MCQ Option Shuffling : No

When drawing large samples from a highly skewed population with an unknown continuous distribution and finite variance, the Central Limit Theorem is the primary theoretical mechanism used to justify standard, asymptotic inference for which of the following?

Options :

1. Inference about population proportions
2. Inference about population means
3. Inference about population variances
4. Inference about population medians

Question Number : 19 Question Id : 4410092108527 Question Type : MCQ Option Shuffling : No

If events X and Y are independent, and $P(X) = 0.4$, $P(Y) = 0.5$, what is the probability that both X and Y do not occur?

Options :

1. 0.2
2. 0.6
3. 0.5
4. 0.3

Question Number : 20 Question Id : 4410092108533 Question Type : MCQ Option Shuffling : No

A fair die is rolled twice. Let S be the sum of the two outcomes. What is the probability that S equals 7?

Options :

1. 0.111
2. 0.167
3. 0.222
4. 0.278

Question Number : 21 Question Id : 4410092108629 Question Type : MCQ Option Shuffling : No

A fair six-sided die is rolled once. If the result is an odd number, a fair coin is flipped once. If the result is an even number, the coin is flipped twice. What is the total number of sample points in this sample space S?

Options :

1. 8
2. 10
3. 18
4. 20

Question Number : 22 Question Id : 4410092108634 Question Type : MCQ Option Shuffling : No

Suppose $Y = aX + b$, where X is a random variable and a, b are constants. How is the expectation of Y related to the expectation of X?

Options :

1. $aE[X] + b$.
2. $abE[X]$
3. $[E[X]]^2 + b$
4. $E[X] + a + b$

Question Number : 23 Question Id : 4410092088767 Question Type : MCQ Option Shuffling : No

In advanced probability theory, if the fourth standardized moment of a symmetrical continuous distribution evaluates precisely to three, what fundamental geometric property is mathematically established regarding its extremities?

Options :

1. It manifests an outlier probability mass precisely equivalent to the standard Gaussian theoretical baseline.
2. It manifests an outlier probability mass strictly superior to the standard Gaussian theoretical baseline.
3. It manifests an outlier probability mass strictly inferior to the standard Gaussian theoretical baseline.
4. It manifests an outlier probability mass entirely orthogonal to the standard Gaussian theoretical baseline.

Question Number : 24 Question Id : 4410092088768 Question Type : MCQ Option Shuffling : No

In quantitative risk modeling, when evaluating the bivariate joint survival function for two strictly independent continuous random variables, which mathematical operation is formally required to determine the precise probability of simultaneous threshold exceedance?

Options :

1. Requires the arithmetic summation of their respective marginal survival probabilities evaluated at the specified thresholds.
2. Requires the sequential minimization of their respective marginal survival probabilities evaluated at the specified thresholds.
3. Requires the absolute subtraction of their respective marginal survival probabilities evaluated at the specified thresholds.
4. Requires the multiplicative product of their respective marginal survival probabilities evaluated at the specified thresholds.

Question Number : 25 Question Id : 4410092108526 Question Type : MCQ Option Shuffling : No

Consider a continuous sample space $S = [0, 1]$. If an event E consists of all points x in S such that $x > 0.7$, what is the probability of E under the uniform probability model?

Options :

1. 0.2
2. 0.3
3. 0.5
4. 0.4

Question Number : 26 Question Id : 4410092108529 Question Type : MCQ Option Shuffling : No

A diagnostic test for a disease has a sensitivity of 95% and specificity of 90%. The disease prevalence in the population is 2%. If a randomly selected individual tests positive, what is the probability that the individual actually has the disease?

Options :

1. 0.162
2. 0.211
3. 0.864
4. 0.190

Question Number : 27 Question Id : 4410092108531 Question Type : MCQ Option Shuffling : No

Suppose a random variable W has mean 5 and variance 9. What is the expectation of $(W - 2)^2$?

Options :

1. 18
2. 9
3. 16
4. 12

Question Number : 28 Question Id : 4410092108632 Question Type : MCQ Option Shuffling : No

A risk manager knows that 8% of loans are non-performing. If a loan is non-performing, there is a 60% chance that it was issued in the last year. Of all loans, 15% were issued in the last year. Then the probability that a randomly selected loan issued in the last year is non-performing is approximately equal to:

Options :

1. 0.48
2. 0.60
3. 0.12
4. 0.32

Question Number : 29 Question Id : 4410092108633 Question Type : MCQ Option Shuffling : No

Suppose X and Y are random variables with joint distribution function as $F(x, y)$. Then

$P(a < X \leq b, Y > c)$ is

Options :

1. $F_X(b) - F_X(a) - F(b, c) + F(a, c)$
2. $F(b, c) - F(a, c)$
3. $F_X(b) - F_X(a)$
4. $F_X(c) - F(a, b)$

Question Number : 30 Question Id : 4410092088777 Question Type : MCQ Option Shuffling : No

What property is preserved when the likelihood is factorized as per the Fisher-Neyman factorization theorem?

Options :

1. All parameter information is retained
2. All statistical estimators are consistent
3. All calculated statistics are unbiased
4. All original data is discarded

Question Number : 31 Question Id : 4410092108647 Question Type : MCQ Option Shuffling : No

In the exponential family, which form does the minimal sufficient statistic typically take for independent observations?

Options :

1. An average of the observations
2. A difference between the largest and smallest values
3. A sum of functions applied to each observation
4. A product of the observations

Question Number : 32 Question Id : 4410092108541 Question Type : MCQ Option Shuffling : No

If $X_1, \dots, X_n$ are i.i.d. $N(\mu, \sigma^2)$ random variables and both μ and σ^2 are unknown, which of the following pairs forms a sufficient statistic for (μ, σ^2) ?

Options :

1. The sample median and sample variance, since median is robust to outliers
2. The sample mean and sample range, since they jointly measure location and dispersion.
3. The sum of X_i and sample median, since both capture location information
4. The sample mean and sum of X_i squared, since they fully determine the likelihood

Question Number : 33 Question Id : 4410092108542 Question Type : MCQ Option Shuffling : No

Given $X_1, X_2, \dots, X_n$ are i.i.d. from a distribution with density $f(x; \theta) = \exp(-(x-\theta))$ for $x \geq \theta$, zero otherwise. Which statistic is sufficient for θ according to the factorization theorem?

Options :

1. The sum of all X_i values
2. The minimum of all X_i values
3. The sample mean of all X_i values
4. The maximum of all X_i values

Question Number : 34 Question Id : 4410092088786 Question Type : MCQ Option Shuffling : No

What is a necessary condition for the Chapman–Robbins–Kiefer lower bound to hold?

Options :

1. Estimator must be efficient
2. Estimator must be unbiased
3. Estimator must be sufficient
4. Estimator must be complete

Question Number : 35 Question Id : 4410092108548 Question Type : MCQ Option Shuffling : No

Given a regular parametric family with likelihood $L(\theta)$, which condition must hold for Fisher information to be a valid measure of information about θ ?

Options :

1. The parameter space must be bounded
2. The density must be independent of θ
3. The support of the density must not depend on θ
4. The likelihood is maximum at the true parameter

Question Number : 36 Question Id : 4410092088781 Question Type : MCQ Option Shuffling : No

Which technique is commonly used to find the UMVUE of a parameter?

Options :

1. Applying the Rao-Blackwell theorem to a sufficient statistic
2. Choosing the estimator with the least squares property
3. Selecting the estimator based on the Cramer-Rao lower bound
4. Using the method of moments estimator directly

Question Number : 37 Question Id : 4410092088783 Question Type : MCQ Option Shuffling : No

Which property ensures that the Lehmann–Scheffe theorem applies to an estimator?

Options :

1. The estimator must be unbiased and sufficient for the parameter.
2. The estimator must be unbiased and consistent for the parameter.
3. The estimator must be a function of a complete but not sufficient statistic.
4. The estimator must be a function of a sufficient and complete statistic.

Question Number : 38 Question Id : 4410092088787 Question Type : MCQ Option Shuffling : No

An analyst is using the method of moments to estimate both the shape (α) and scale (β) parameters of a Gamma distribution from a given random sample. To successfully construct and resolve the required system of equations, what is the minimum number of distinct sample moments the analyst must compute?

Options :

1. Exactly two sample moments
2. Exactly four sample moments
3. Exactly three sample moments
4. Exactly one sample moment

Question Number : 39 Question Id : 4410092108545 Question Type : MCQ Option Shuffling : No

A researcher claims that the sample median is always an unbiased estimator for the population median, regardless of the sample size or underlying distribution. Which statement below best evaluates this claim?

Options :

1. The claim is unconditionally valid under standard regularity conditions of parametric families
2. The claim fails exclusively when the underlying distribution possesses symmetric heavy tails
3. The claim is mathematically true only when evaluating perfectly continuous normal families
4. The claim is demonstrably false due to inherent bias within skewed distributions

Question Number : 40 Question Id : 4410092108652 Question Type : MCQ Option Shuffling : No

When is the Fisher information function for a parameter zero?

Options :

1. When the likelihood is differentiable but non-constant
2. When the likelihood does not depend on the parameter
3. When the parameter is not estimable but the likelihood depends on it
4. When the observed data have high variance

Question Number : 41 Question Id : 4410092108653 Question Type : MCQ Option Shuffling : No

In which case does an unbiased estimator fail to achieve the Cramér–Rao lower bound?

Options :

1. The estimator does not utilize all the information available in the data.
2. The estimator is a function of a sufficient statistic.
3. The estimator is derived from a regular family of distributions.
4. The estimator is obtained by the method of maximum likelihood.

Question Number : 42 Question Id : 4410092108655 Question Type : MCQ Option Shuffling : No

A researcher applies the method of moments to a distribution with skewness and kurtosis. What challenge may arise in estimating all parameters using this method?

Options :

1. Large sample sizes always guarantee unique estimators
2. Using only the first two moments always suffices for all parameters
3. Additional conditions always solve identifiability problems
4. The sample moments may not uniquely determine the population parameters

Question Number : 43 Question Id : 4410092108656 Question Type : MCQ Option Shuffling : No

Suppose the likelihood function for a parameter is strictly concave and twice differentiable. What does this imply about the maximum likelihood estimator?

Options :

1. There may be multiple estimators each maximizing the likelihood
2. The estimator is undefined due to the likelihood function's shape
3. The estimator can be any value that increases the likelihood
4. The estimator is unique and determined by zero of the score function

Question Number : 44 Question Id : 4410092108659 Question Type : MCQ Option Shuffling : No

Which option best explains why randomized tests are constructed in hypothesis testing?

Options :

1. To attain an exact significance level
2. To guarantee unbiased decisions
3. To maximize the likelihood ratio
4. To always increase the test's power

Question Number : 45 Question Id : 4410092088795 Question Type : MCQ Option Shuffling : No

When analyzing a sample of paired differences utilizing the Wilcoxon signed-rank test, what is the minimum theoretical measurement scale required for those specific differences?

Options :

1. Requires an ordinal scale to meaningfully rank the differences.
2. Requires a nominal scale to simply categorize the differences
3. Requires an interval scale to exactly average the differences
4. Requires a ratio scale to correctly divide the differences

Question Number : 46 Question Id : 4410092108556 Question Type : MCQ Option Shuffling : No

Suppose you are testing a simple null hypothesis against a simple alternative using a test function with a critical region of size 0.05. If the test is randomized on the boundary of the critical region, which of the following statements is correct?

Options :

1. The probability of Type I error equals 0.05 due to randomization at the boundary.
2. The probability of Type I error is unaffected by randomization at the boundary.
3. The probability of Type I error can be less than 0.05 if the boundary is included.
4. The probability of Type I error exceeds 0.05 if the test is randomized.

Question Number : 47 Question Id : 4410092088792 Question Type : MCQ Option Shuffling : No

How does randomization in tests affect the interpretation of p-values?

Options :

1. It always decreases the p-value
2. It introduces a probability component to the test decision at the critical value
3. It removes all uncertainty from p-value interpretation
4. It causes the p-value to be undefined

Question Number : 48 Question Id : 4410092088794 Question Type : MCQ Option Shuffling : No

In the context of hypothesis testing, what does 'nested models' mean for the likelihood ratio test?

Options :

1. The models have overlapping but not nested parameters.
2. The models have unrelated parameter spaces.
3. The models are always disjoint in parameter space.
4. The null model is a special case of the alternative model.

Question Number : 49 Question Id : 4410092108559 Question Type : MCQ Option Shuffling : No

A researcher tests $H_0: \mu = 0$ versus $H_1: \mu \neq 0$ in a normal model with known variance, using a sample mean of 0.85 from 100 observations and variance 1. What is the asymptotic value of the LRT statistic?

Options :

1. 85
2. 42.5
3. 170
4. 72.25

Question Number : 50 Question Id : 4410092108665 Question Type : MCQ Option Shuffling : No

Which sampling method divides the population into distinct groups before selecting samples from each group?

Options :

1. Simple random sampling
2. Systematic sampling
3. Cluster sampling
4. Stratified sampling

Question Number : 51 Question Id : 4410092088799 Question Type : MCQ Option Shuffling : No

In Probability Proportional to Size sampling, what does 'size measure' commonly refer to?

Options :

1. An auxiliary variable indicating the relative magnitude or importance of each unit.
2. The physical dimensions of each unit recorded during the sampling process.
3. The predetermined sample size specified before drawing the units.
4. The cumulative total of all units selected during the sampling procedure.

Question Number : 52 Question Id : 4410092108562 Question Type : MCQ Option Shuffling : No

Which scenario demonstrates a potential pitfall when using systematic sampling instead of stratified sampling in a heterogeneous population?

Options :

1. Systematic sampling always ensures proportional representation of all subgroups
2. Stratified sampling introduces bias by forcing subgroup representation
3. Randomizing the start point in systematic sampling eliminates subgroup overrepresentation
4. Systematic sampling may miss subgroups if periodicity aligns with subgroup distribution

Question Number : 53 Question Id : 4410092088801 Question Type : MCQ Option Shuffling : No

In regression estimation, what happens if the auxiliary variable is poorly measured?

Options :

1. The estimator's accuracy improves
2. Efficiency always increases
3. Reliability remains unaffected
4. The regression estimator becomes less reliable

Question Number : 54 Question Id : 4410092108566 Question Type : MCQ Option Shuffling : No

In survey sampling, suppose the auxiliary variable X has a known population mean $\bar{X}$, and the sample shows a non-linear relation between Y and X with a non-zero intercept. Let $\bar{Y}^R$ denote the ratio estimator, $\bar{Y}^{reg}$ the regression estimator, and $\bar{y}$ the simple sample mean. Which statement correctly describes their large-sample properties?

Options :

1. The ratio estimator remains perfectly unbiased in finite samples, while the regression estimator introduces bias due to the intercept term
2. The regression estimator achieves lower asymptotic variance than the ratio estimator as it minimises residual variation with biasness
3. The simple sample mean always outperforms the ratio estimator whenever the regression slope is less than half the population ratio $\bar{Y}/\bar{X}$
4. The regression estimator and ratio estimator have identical variance only when the correlation between Y and X equals zero

Question Number : 55 Question Id : 4410092108666 Question Type : MCQ Option Shuffling : No

Why might stratified sampling yield greater precision in estimating population parameters than systematic sampling in a heterogeneous population?

Options :

1. Because it relies on random periodic selection to improve estimates
2. Because it minimizes the cost of collecting population data
3. Because it increases the total sample size taken from the population
4. Because it controls variance within each subgroup and aggregates across them

Question Number : 56 Question Id : 4410092088805 Question Type : MCQ Option Shuffling : No

What is the primary statistical effect of randomizing the treatment assignments within the restricted framework of a Latin square design?

Options :

1. It effectively reduces the introduction of systematic bias
2. It entirely eliminates the requirement for experimental blocking
3. It completely isolates the interactions between nuisance variables
4. It mathematically decreases the total experimental error variance

Question Number : 57 Question Id : 4410092108571 Question Type : MCQ Option Shuffling : No

Which one of the following best describes the main effect of factor A in a 2^2 factorial experiment?

Options :

1. The response observed at the highest level of factor A
2. The average change in response when factor A changes from low to high
3. The average response when factor B is held constant
4. The change in response when both factors change simultaneously

Question Number : 58 Question Id : 4410092088807 Question Type : MCQ Option Shuffling : No

Which condition increases the likelihood of confounding in a 2^k factorial experiment?

Options :

1. Reducing the number of experimental runs below the full factorial requirement
2. Increasing the number of replicates per combination
3. Testing all possible combinations of factor levels
4. Removing all factors except one from the experiment

Question Number : 59 Question Id : 4410092108670 Question Type : MCQ Option Shuffling : No

What is the main statistical advantage of a completely randomized design?

Options :

1. It minimizes bias through random allocation
2. It increases sample size for each group
3. It increases external validity of findings
4. It maximizes differences between treatments

Question Number : 60 Question Id : 4410092108676 Question Type : MCQ Option Shuffling : No

In analyzing a full factorial design, what is an advantage over confounded models?

Options :

1. Only the main effects can be estimated independently.
2. The statistical interpretation is limited strictly to partial effects.
3. All main and interaction effects can be estimated independently.
4. Confounding increases for higher-order effects.

Question Number : 61 Question Id : 4410092088809 Question Type : MCQ Option Shuffling : No

Why are full factorial models rarely used in large experiments?

Options :

1. Full models ensure unbiased estimation for every possible effect
2. Full models require a large number of experimental runs, making them resource-intensive and impractical
3. Full models reduce the experimental error variance compared to confounded models
4. Full models guarantee smaller block sizes for all experiments

Question Number : 62 Question Id : 4410092108674 Question Type : MCQ Option Shuffling : No

During the analysis of a Latin square experiment, what statistical model component specifically accounts for the nuisance variables?

Options :

1. Nuisance variables are combined with treatment effects in the model
2. Nuisance variables are included as fixed effects in the model
3. Effects of nuisance variables are ignored during the analysis
4. Nuisance variables are treated as random effects in the analysis

Question Number : 63 Question Id : 4410092108675 Question Type : MCQ Option Shuffling : No

If an interaction effect between two factors in a 2^k factorial experiment is significant, what does this imply?

Options :

1. The factors are statistically independent
2. Both main effects must be insignificant
3. Only one factor influences the response
4. One factor's effect depends on the other factor

Question Number : 64 Question Id : 4410092108677 Question Type : MCQ Option Shuffling : No
If a confounded ANOVA table has:

Source	Degrees of freedom
Blocks	1
Treatments	30

Then the experiment is:

Options :

1. 2^5 in two blocks
2. 2^5 in four blocks
3. 2^6 in two blocks
4. 2^5 in three blocks

Question Number : 65 Question Id : 4410092108681 Question Type : MCQ Option Shuffling : No
For which type of state, do the limiting n-step transition probabilities equal zero in the long run?

Options :

1. Transient state
2. Irreducible state
3. Recurrent state
4. Aperiodic state

Question Number : 66 Question Id : 4410092088817 Question Type : MCQ Option Shuffling : No
In the context of stochastic modeling, how does increasing the number of simulation runs affect the estimation of long-run state frequencies?

Options :

1. It ensures that empirical state frequencies strictly converge toward the theoretical stationary distribution.
2. It ensures that inherent stochastic variance within the underlying transition process completely disappears.
3. It ensures that the foundational transition matrix dynamically adapts to eliminate transient behavior.
4. It ensures that statistical bias introduced by the initial starting state progressively amplifies.

Question Number : 67 Question Id : 4410092108579 Question Type : MCQ Option Shuffling : No
For a two-state Markov chain with transition matrix $P = \begin{bmatrix} 0.7 & 0.3 \\ 0.6 & 0.4 \end{bmatrix}$. Derive the stationary distribution.

Options :

1. 0.4, 0.6
2. 0.6, 0.4
3. 0.5, 0.5
4. 0.67, 0.33

Question Number : 68 Question Id : 4410092108576 Question Type : MCQ Option Shuffling : No

Consider a Markov chain on the integers $\{0, 1, \dots, N\}$ with transition probabilities $p_{\{n, n+1\}} = p$ and $p_{\{n, n-1\}} = 1-p$ for $0 < n < N$, where $0 < p < 1$. The states 0 and N are absorbing barriers. Let P_i denote the probability of absorption at state N starting from state i ($0 < i < N$). Which of the following statements is true regarding the curvature and asymptotic behavior of P_i ?

Options :

1. For any fixed i, if $p > 1/2$, P_i is a strictly concave function of i, and the limit of P_i as N approaches infinity is 1.
2. For any fixed i, if $p < 1/2$, P_i is a strictly convex function of i, and the limit of P_i as N approaches infinity is 0.
3. For $p = 1/2$, P_i is linear in i, and its value remains strictly dependent on N as N approaches infinity for all fixed $i > 0$.
4. D. If $p > 1/2$, the sequence P_i is strictly convex with respect to i, and the limit of P_i as N approaches infinity is $1 - ((1-p)/p)^i$. [1,2,3]

Question Number : 69 Question Id : 4410092108582 Question Type : MCQ Option Shuffling : No

Which of the following is a valid reason for poor convergence to the stationary distribution in Markov chain simulation?

Options :

1. The chain contains periodic states with period greater than one.
2. The chain maintains an irreducible and strictly aperiodic structure.
3. The chain possesses a strictly unique stationary distribution limit.
4. The chain lacks strict reversibility despite a stationary structure.

Question Number : 70 Question Id : 4410092108682 Question Type : MCQ Option Shuffling : No

If a Markov chain with a finite state space has a unique stationary distribution, what does this imply about the limiting behaviour of its n-step transition probabilities?

Options :

1. They become periodic with period greater than one
2. They converge to the stationary distribution
3. They diverge depending on the starting state
4. They oscillate between multiple distributions

Question Number : 71 Question Id : 4410092108685 Question Type : MCQ Option Shuffling : No

A Markov chain is simulated for a system with rare transitions between states. Which adjustment will help ensure accurate estimation of rare-event probabilities?

Options :

1. Artificially alter the transition matrix without applying statistical re-weighting.
2. Ignore the rare state transitions entirely during the primary statistical analysis.
3. Reduce the total simulation steps to focus exclusively on the highly probable events.
4. Extend the total number of simulation steps substantially to capture the rare events.

Question Number : 72 Question Id : 4410092108583 Question Type : MCQ Option Shuffling : No

Given a linear model $Y = X\beta + \varepsilon$ with $X = [[1,1],[1,1],[1,1]]$, which of the following is estimable?

Options :

1. β_2 alone
2. β_1 alone
3. The difference $\beta_1 - \beta_2$
4. The sum $\beta_1 + \beta_2$

Question Number : 73 Question Id : 4410092108584 Question Type : MCQ Option Shuffling : No

Which statement about the width of a confidence interval for the mean response at a given value of the predictor in a linear model is most accurate?

Options :

1. The width remains constant with larger sample size
2. The width is unaffected by response variability
3. The width increases with larger sample size, all else equal
4. The width decreases as sample size increases, all else equal

Question Number : 74 Question Id : 4410092108687 Question Type : MCQ Option Shuffling : No

Under which condition does the Gauss-Markov theorem guarantee that the OLS estimator is the BLU estimator?

Options :

1. The error terms exhibit an unequal variance.
2. The error terms follow a normal distribution.
3. The error terms are zero-mean, uncorrelated, and homoscedastic.
4. The predictor variables are strictly independent of each other.

Question Number : 75 Question Id : 4410092108688 Question Type : MCQ Option Shuffling : No

A researcher tests multiple linear hypotheses on regression coefficients. What is a potential risk if each test is performed at the same significance level without adjustment?

Options :

1. All regression models inherently become unbiased.
2. The individual chance of a type II error decreases.
3. The overall probability of a type I error increases.
4. All genuinely true hypotheses are always accepted.

Question Number : 76 Question Id : 4410092088820 Question Type : MCQ Option Shuffling : No

When constructing a confidence interval for the mean response at a new value of the predictor in a linear model, what additional source of uncertainty must be included compared to predicting the mean at observed data points?

Options :

1. The variability in the new predictor value's estimate
2. The sample size adjustment for the new observation
3. The estimation of the error variance only
4. The fit of the regression model to the existing data

Question Number : 77 Question Id : 4410092108588 Question Type : MCQ Option Shuffling : No

A researcher fits a logistic regression model and observes that the estimated probability for a specific observation is 0.01. Which of the following best describes the behavior of the log-likelihood function for this observation when the true outcome is 1?

Options :

1. The log-likelihood is undefined
2. The log-likelihood is close to zero
3. The log-likelihood is positive
4. The log-likelihood is a large negative value

Question Number : 78 Question Id : 4410092108693 Question Type : MCQ Option Shuffling : No

What condition must the matrix A satisfy for the quadratic form $x'Ax$ to follow a chi-squared distribution?

Options :

1. The matrix A must be symmetric and idempotent
2. The matrix A must be strictly non-singular
3. The matrix A must be strictly skew-symmetric
4. The matrix A must be diagonal and positive

Question Number : 79 Question Id : 4410092108694 Question Type : MCQ Option Shuffling : No

Which assumption is necessary for linear discriminant analysis to yield optimal results?

Options :

1. Each defined group exhibits a multivariate normal distribution.
2. All independent predictor variables are strictly categorical.
3. All defined groups possess exactly equal sample sizes.
4. All independent predictor variables are mutually independent.

Question Number : 80 Question Id : 4410092108692 Question Type : MCQ Option Shuffling : No

Suppose X is a three-dimensional multivariate normal vector with mean μ and covariance Σ . If you observe that two components of X are perfectly correlated, what can be inferred about the determinant of Σ ?

Options :

1. The covariance matrix Σ must be strictly diagonal.
2. The determinant of Σ is exactly equal to zero.
3. The determinant of Σ is mathematically maximized
4. The determinant of Σ is minimized but non zero

Question Number : 81 Question Id : 4410092108695 Question Type : MCQ Option Shuffling : No

In canonical correlation, which method is most suitable for testing the significance of each canonical correlation coefficient?

Options :

1. Bartlett's test for sphericity on correlation matrices
2. Significance testing using Wilks' lambda statistic
3. Bonferroni correction for multiple pairwise comparisons
4. Hotelling's T-square test for group mean differences

Question Number : 82 Question Id : 4410092108697 Question Type : MCQ Option Shuffling : No

What does a k-out-of-n system require for proper operation?

Options :

1. At least k components must function among n total components.
2. Exactly k components must remain functional throughout operation.
3. All n components must function simultaneously at all times.
4. Exactly n minus k components must remain functional to operate.

Question Number : 83 Question Id : 4410092108593 Question Type : MCQ Option Shuffling : No

Suppose 15 identical devices are tested under Type II censoring, and the test stops after 5 failures. The observed failure times are 20, 35, 40, 50, and 60 hours; the remaining 10 devices are censored at 60 hours. Estimate the exponential mean lifetime using the maximum likelihood approach.

Options :

1. The estimator is $(20 + 35 + 40 + 50 + 60 + 10 \times 60) / 5$ hours
2. The estimator is $(10 \times 60) / 15$ hours
3. The estimator is $(20 + 35 + 40 + 50 + 60) / 15$ hours
4. The estimator is $(20 + 35 + 40 + 50 + 60) / 5$ hours

Question Number : 84 Question Id : 4410092088829 Question Type : MCQ Option Shuffling : No

In a reliability study, 20 identical devices are tested, and the test is terminated after the first 7 failures or at 2000 hours, whichever comes first. What type of censoring is exemplified in this scenario?

Options :

1. This is an example of hybrid censoring, as the test ends by combining time and failure criteria.
2. This is an example of Type II censoring, as the test ends after a predetermined number of failures.
3. This is an example of Type I censoring, as the test ends at a fixed time.
4. This is an example of random censoring, as the test ends at random intervals.

Question Number : 85 Question Id : 4410092088831 Question Type : MCQ Option Shuffling : No

Which of the following is a required assumption of linear programming?

Options :

1. Variables must always be integers
2. Duality in every linear programming problem
3. Non-negativity in the objective function coefficients
4. Proportionality in the relationships between variables and constraints

Question Number : 86 Question Id : 4410092108596 Question Type : MCQ Option Shuffling : No

Which of the following statements about the relationship between the primal and dual in linear programming is correct when both are feasible and bounded?

Options :

1. The primal optimal value is less than the dual only when both are unbounded.
2. The optimal value of the dual always exceeds that of the primal.
3. The optimal values of both problems are equal under strong duality.
4. Dual feasibility is not required for equality of optimal values.

Question Number : 87 Question Id : 4410092108599 Question Type : MCQ Option Shuffling : No

In continuous review inventory models with backordering allowed, increasing the backorder cost parameter while keeping other factors constant will most likely result in which effect?

Options :

1. Unchanged reorder points but reduced batch sizes
2. Lower reorder points and reduced average inventory levels
3. Reduced batch sizes and unchanged average inventory
4. Higher reorder points and increased average inventory levels

Question Number : 88 Question Id : 4410092108702 Question Type : MCQ Option Shuffling : No

A hospital emergency room is modeled as an M/M/c system. While keeping arrival and service rates unchanged, what impact does increasing the number of servers have?

Options :

1. Arrival rate in the system increases
2. Service rate per server decreases
3. Queue length increases due to splitting
4. Average waiting time in the queue decreases

Question Number : 89 Question Id : 4410092108704 Question Type : MCQ Option Shuffling : No

Which of the following is a method for removing trend from a time series?

Options :

1. Plotting the data
2. Calculating seasonal indices
3. Calculating moving averages
4. Differencing the data values

Question Number : 90 Question Id : 4410092108600 Question Type : MCQ Option Shuffling : No

Suppose you observe a time series where the ACF and PACF both show significant values at multiple lags without clear cutoff points. What is the best interpretation regarding model identification?

Options :

1. A deterministic trend is present
2. A pure MA model is indicated
3. A pure AR model is indicated
4. An ARMA model may be suitable

Question Number : 91 Question Id : 4410092088836 Question Type : MCQ Option Shuffling : No

What does a PACF with a significant spike only at lag 3 and zeros elsewhere suggest?

Options :

1. ARMA(1,2) process
2. AR(3) process
3. White noise process
4. MA(3) process

Question Number : 92 Question Id : 4410092088838 Question Type : MCQ Option Shuffling : No

When applying the Holt–Winters multiplicative model, what is the consequence of using a seasonal index close to zero?

Options :

1. A near-zero seasonal index will lead to no effect on the resulting forecast
2. A near-zero seasonal index will drastically reduce the forecasted value during that season
3. A near-zero seasonal index will amplify the trend component in the forecast
4. A near-zero seasonal index will stabilize the forecasted value across all periods

Question Number : 93 Question Id : 4410092108603 Question Type : MCQ Option Shuffling : No

In the context of white noise testing, which property must be true for a time series to be classified as pure white noise?

Options :

1. All autocorrelations are exactly zero at all nonzero lags
2. The autocorrelation function decays exponentially to zero
3. The variance of the observations increases steadily over time
4. The mean of the series follows a deterministic seasonal pattern

Question Number : 94 Question Id : 4410092088842 Question Type : MCQ Option Shuffling : No

Which group is primarily responsible for the execution and day-to-day leadership of cross-functional Six Sigma projects within an organization?

Options :

1. Certified Six Sigma Green Belts
2. Executive Champions and Project Sponsors
3. Corporate Quality Assurance Management Teams
4. Certified Six Sigma Black Belts

Question Number : 95 Question Id : 4410092088845 Question Type : MCQ Option Shuffling : No

How does increasing the sample size affect producer risk, assuming all else remains constant?

Options :

1. Increasing the sample size does not affect producer risk.
2. Increasing the sample size generally reduces producer risk.
3. Increasing the sample size makes producer risk random.
4. Increasing the sample size generally increases producer risk.

Question Number : 96 Question Id : 4410092108605 Question Type : MCQ Option Shuffling : No

A process control table lists multiple process parameters measured every hour. If an outlier is detected in one parameter at a specific time, what is the most logical initial step using the tabular data?

Options :

1. Recalculate the summary statistics for the table
2. Check all parameters for anomalies at that time
3. Compare that parameter's values at neighboring time intervals
4. Immediately recalibrate the measurement equipment

Question Number : 97 Question Id : 4410092088841 Question Type : MCQ Option Shuffling : No

What is a recommended approach when designing a tabular form for multi-step process control?

Options :

1. Use a single column to combine all process steps
2. Merge process steps to reduce number of table columns
3. Omit stepwise details to simplify the table
4. Keep each process step in a separate column with entries.

Question Number : 98 Question Id : 4410092088846 Question Type : MCQ Option Shuffling : No

How does a double sampling plan help optimize inspection resources in a manufacturing process?

Options :

1. It can reduce the average number of items inspected by making early decisions when possible.
2. It inspects the same number of items regardless of sample outcomes.
3. It requires the maximum possible inspection for every lot.
4. It always requires more inspections than single sampling plans regardless of results.

Question Number : 99 Question Id : 4410092108708 Question Type : MCQ Option Shuffling : No

In a CUSUM chart, what would be the most probable reason for consistent false alarms during routine process monitoring?

Options :

1. Parameter selection based on process specifications
2. Poorly chosen reference value and decision threshold
3. Process operating within normal stable conditions
4. Excessively large sample size per plotted point

Question Number : 100 Question Id : 4410092108711 Question Type : MCQ Option Shuffling : No

If a process capability analysis yields $C_p = 0.8$ and $C_{pk} = 0.8$, what is the best course of action for process improvement?

Options :

1. Increase sample size for more accurate capability estimates.
2. Reduce process variability to enhance both capability and centering.
3. Shift the process mean toward the target value to improve centering.
4. Change specification limits to accommodate current process spread.